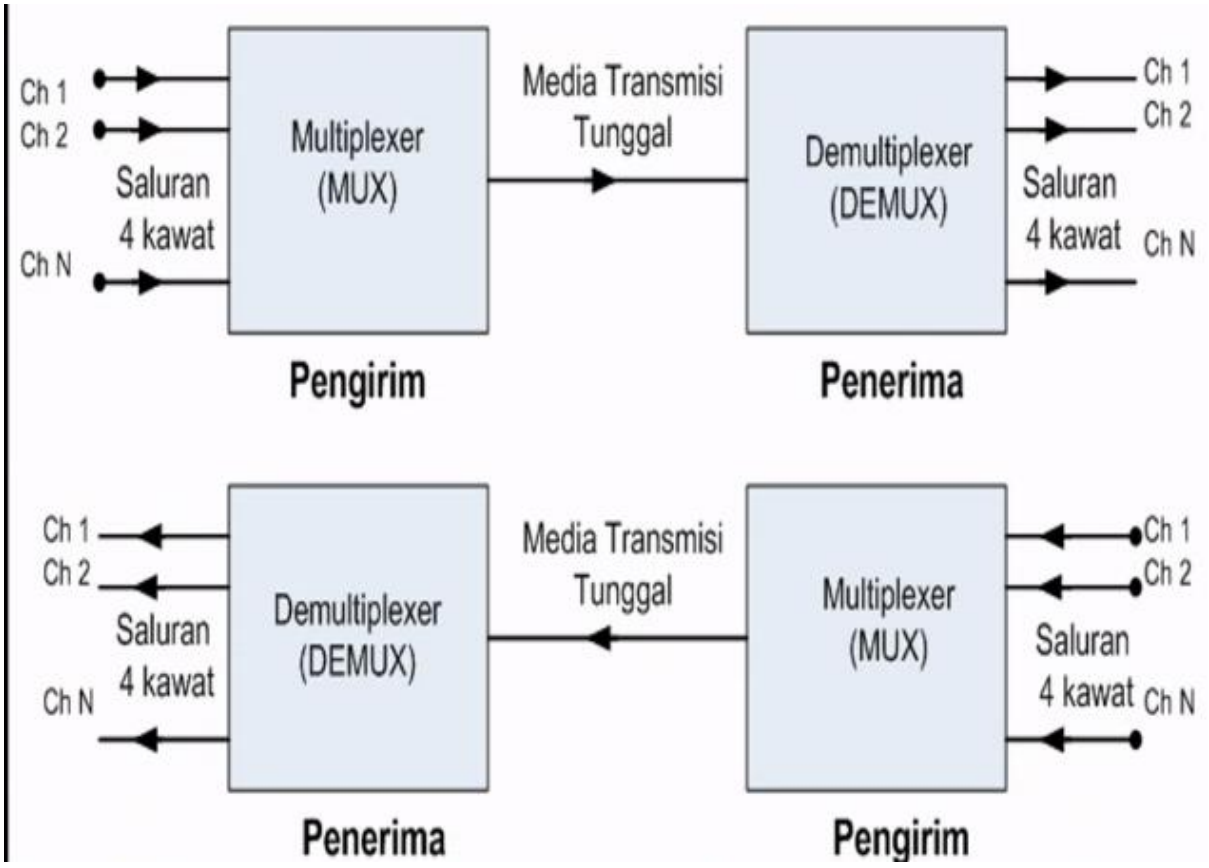
Multiplexing and Demultiplexing (Channelization)

➢ Multiplexing:

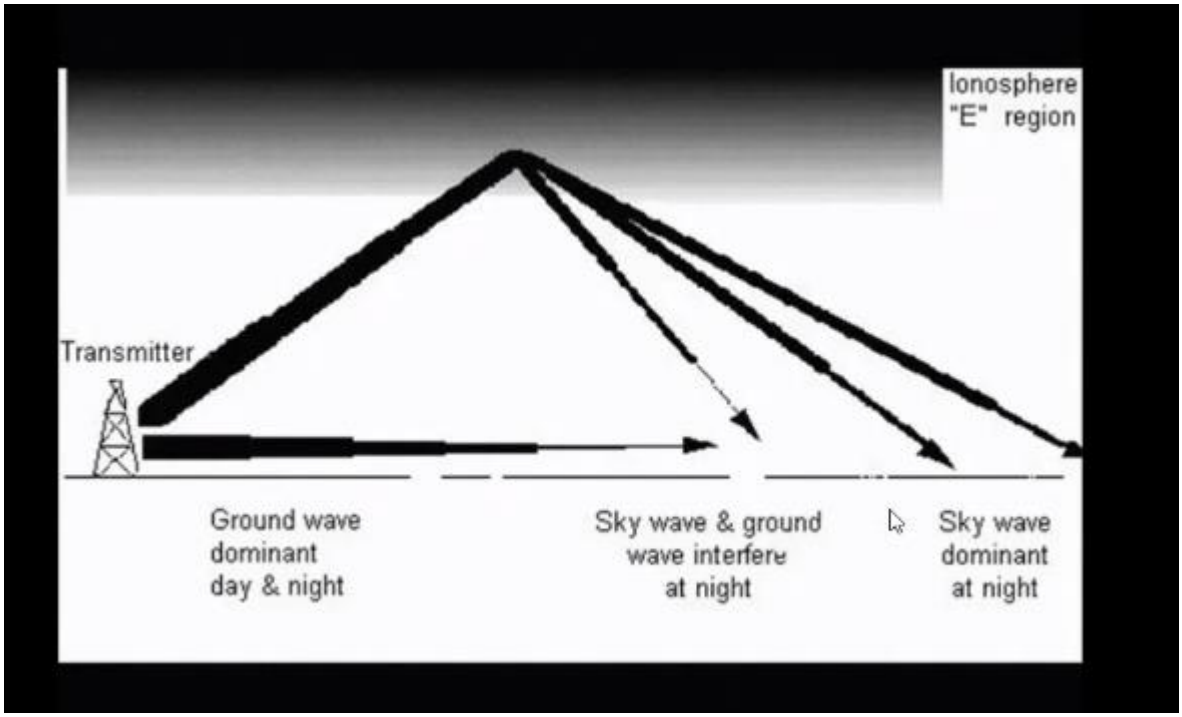
- The **combining of two or more information channels** onto a **common transmission medium.**
- **Two or more** simultaneous **transmissions** on a **single circuit**
- **Multiplexing:** name given to **techniques** which allow **more than one message** to be transferred via the **same communi-cation channel.**
- **Multiplexing** are **techniques** which allow **k users** to **occupy a channel** for the duration in time, that the channel is available.

- **Multiplexing** to refer to the **combination** of information streams from multiple sources for transmission over a ***shared medium***
- Multiplexing adalah transmisi dua atau lebih kanal (Tele)komunikasi secara serentak melalui satu media transmisi.
- **Multiplexor (MUX)** is a mechanism that implements the concept.
- **Demultiplexing** to refer to the **separation** of a combination back into separate information streams
- **Demultiplexor (DEMUX)** to refer to a mechanism that implements the concept



- **Media Transmisi Tunggal (Shared Medium):** could be a transmission line;

- Twisted pair
- Co-axial cable
- A radio system
- Fibre optic cable
- Etc.



Sistem Televisi

❖ **Pemancar TV, dua sinyal yang dipancarkan sekaligus :**

1. Sinyal gambar yang dimodulasi amplitudo (AM-VSB)
2. Sinyal suara yang dimodulasi frekuensi (FM).

❖ **Kedua sinyal dibangkitkan sbg. frekuensi IF, rec. CCIR :**

1. Frekuensi sinyal pembawa gambar sebesar 38,9 MHz d
2. Frekuensi sinyal pembawa suara 33,4 MHz.
3. Frekuensi kedua sinyal tsb. digeser ke frekuensi kerjanya sesuai dengan nomor saluran kanal yang dikehendaki.

❖ **Frekuensi kerja Pemancar TV berada pada spektrum :**

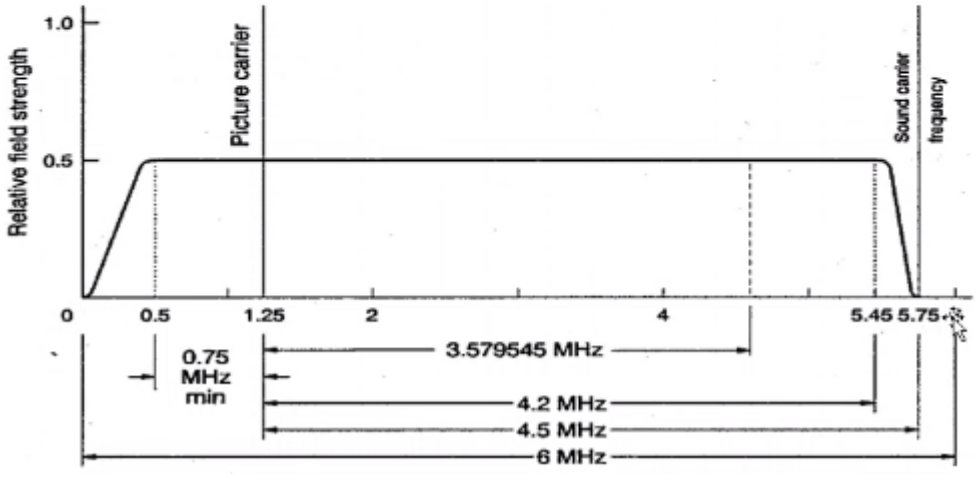
1. VHF (174 - 230 MHz) dan
2. UHF (470 - 806 MHz).

❖TV Broadcast Channels regulated by FCC

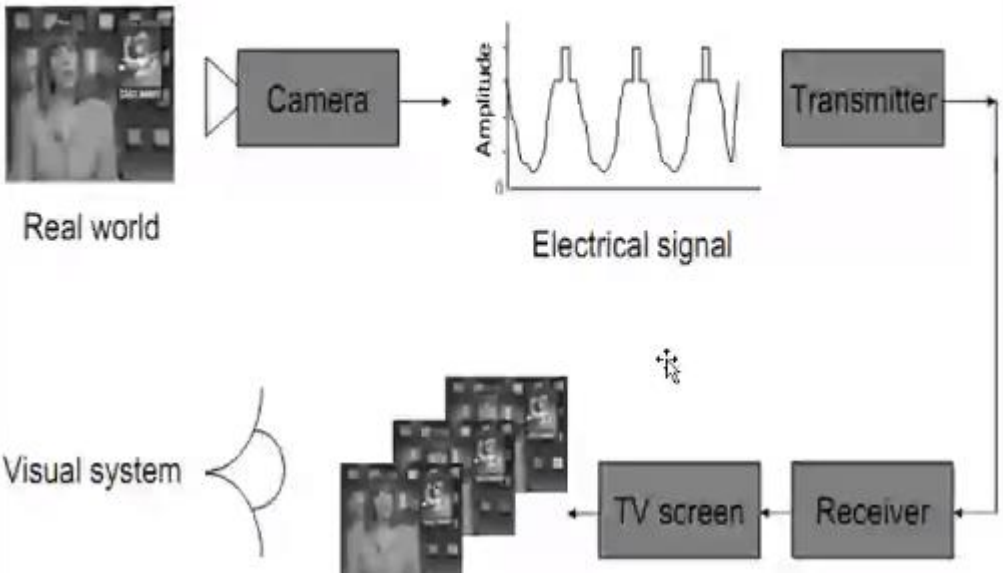
Channel	Frequency, MHz	Channel	Frequency
Low-Band VHF		UHF (cont.)	
2	54–60	35	596–602
3	60–66	36	602–608
4	66–72	37	608–614
5	76–82	38	614–620
6	82–88	39	620–626
High-Band VHF		40	626–632
7	174–180	41	632–638
8	180–186	42	638–644
9	186–192	43	644–650
10	192–198	44	650–656
11	198–204	45	656–662
12	204–210	46	662–668
13	210–216	47	668–674
UHF		48	674–680
14	470–476	49	680–686
15	476–482	50	686–692
16	482–488	51	692–698
17	488–494	52	698–704
18	494–500	53	704–710
19	500–506	54	710–716
20	506–512	55	716–722
21	512–518	56	722–728
22	518–524	57	728–734
23	524–530	58	734–740
24	530–536	59	740–746
25	536–542	60	746–752
26	542–548	61	752–758
27	548–554	62	758–764
28	554–560	63	764–770
29	560–566	64	770–776
30	566–572	65	776–782
31	572–578	66	782–788
32	578–584	67	788–794
33	584–590	68	794–800
34	590–596	69	800–806

❖Bandwidth Sinyal TV

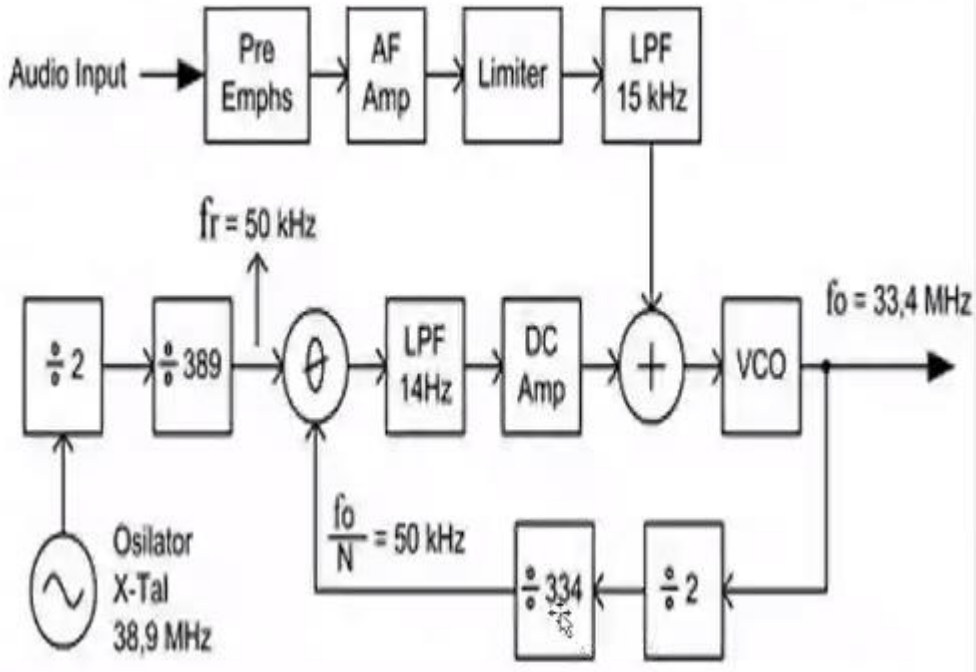
It has two carriers, the visual carrier 1.25 MHz above the lower channel edge and the aural carrier 4.5 MHz above the visual. The visual carrier is amplitude-modulated by the video signal. The aural carrier is frequency-modulated with a pick deviation of ±25kHz (compared to ±75kHz for FM broadcasting).



❖ Basic TV system

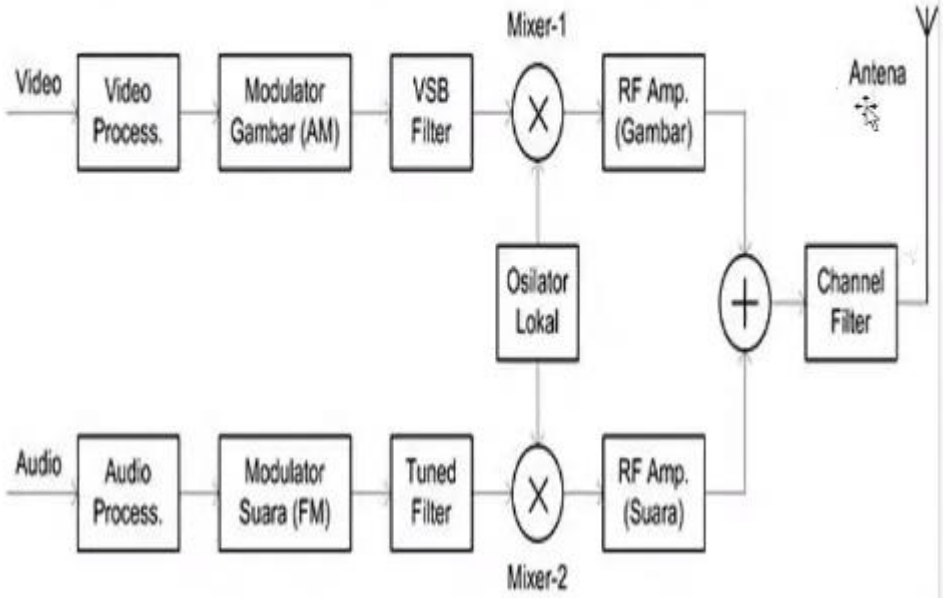


❖ Modulator FM untuk TV

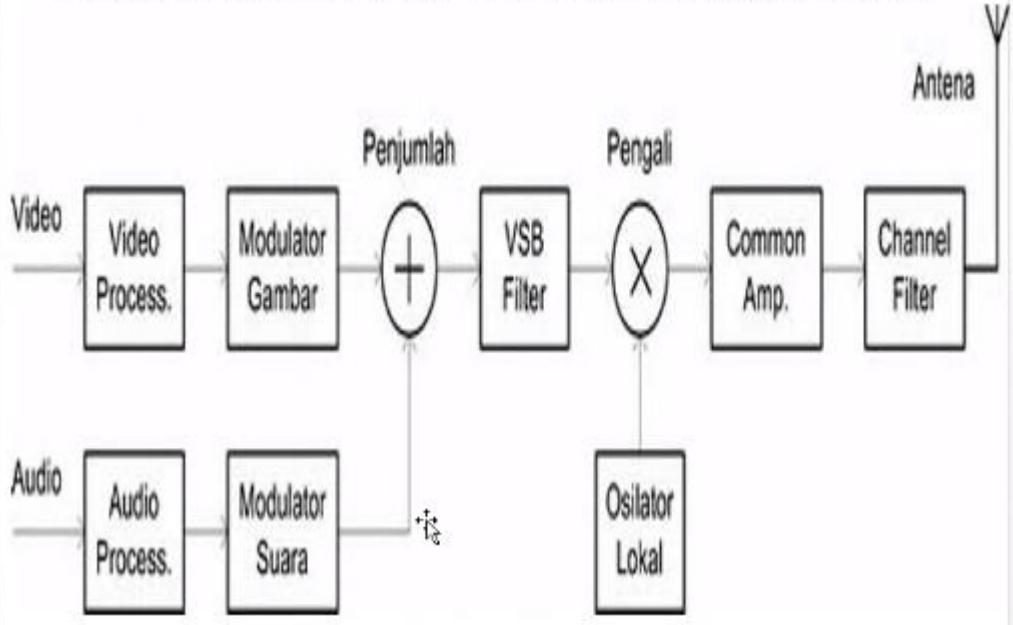


❖ Prinsip Pemancar TV

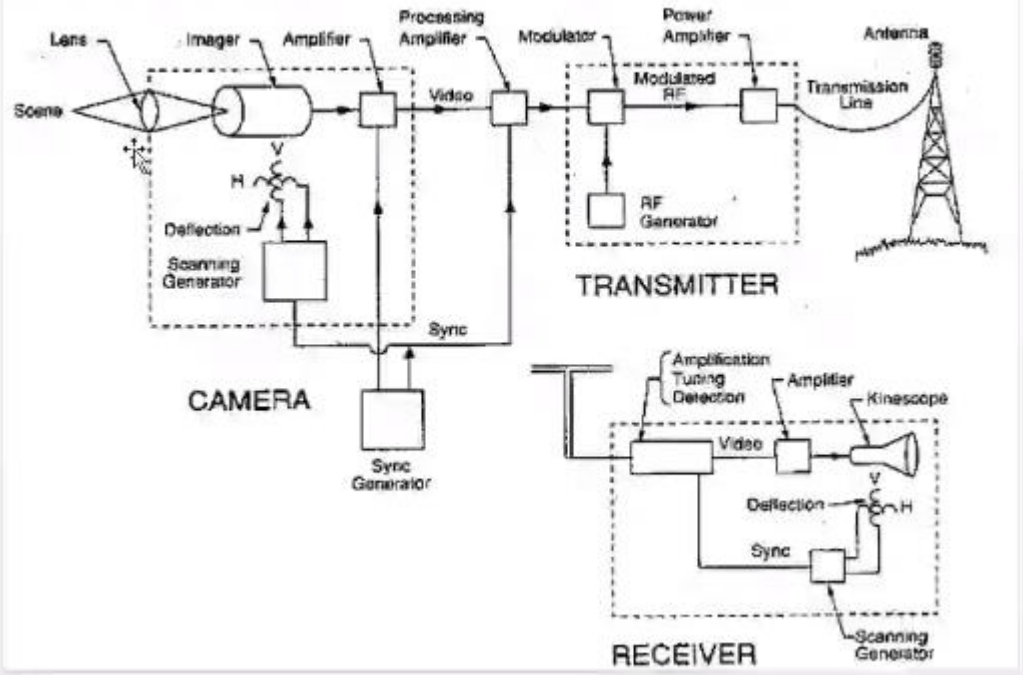
- Diagram Blok Pemancar TV dengan Penguat Terpisah



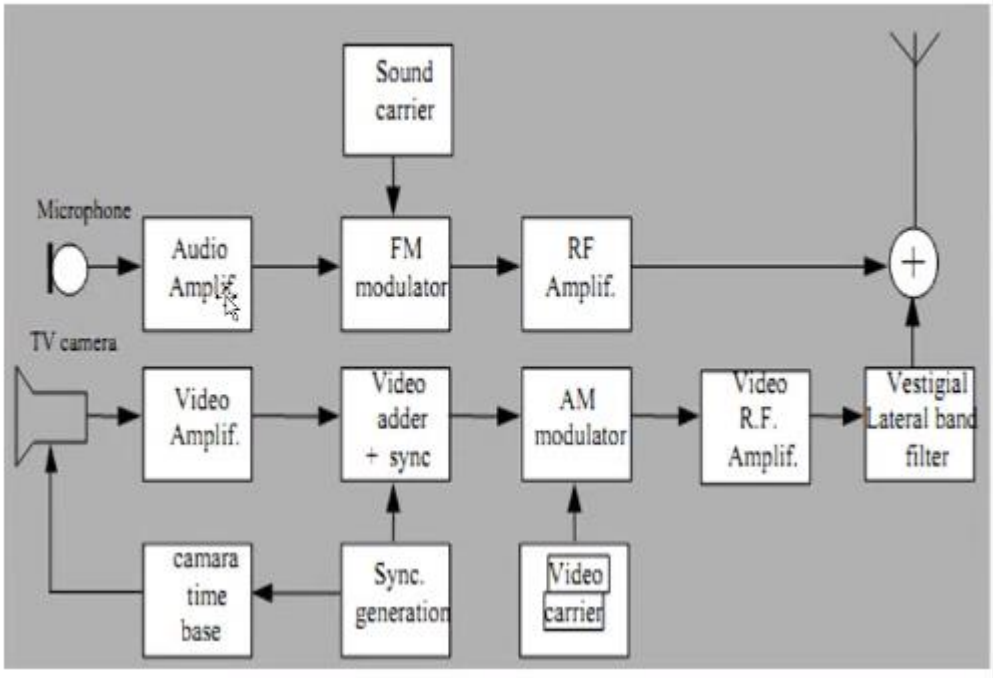
- **Diagram Blok Pemancar TV dengan Penguat Bersama**



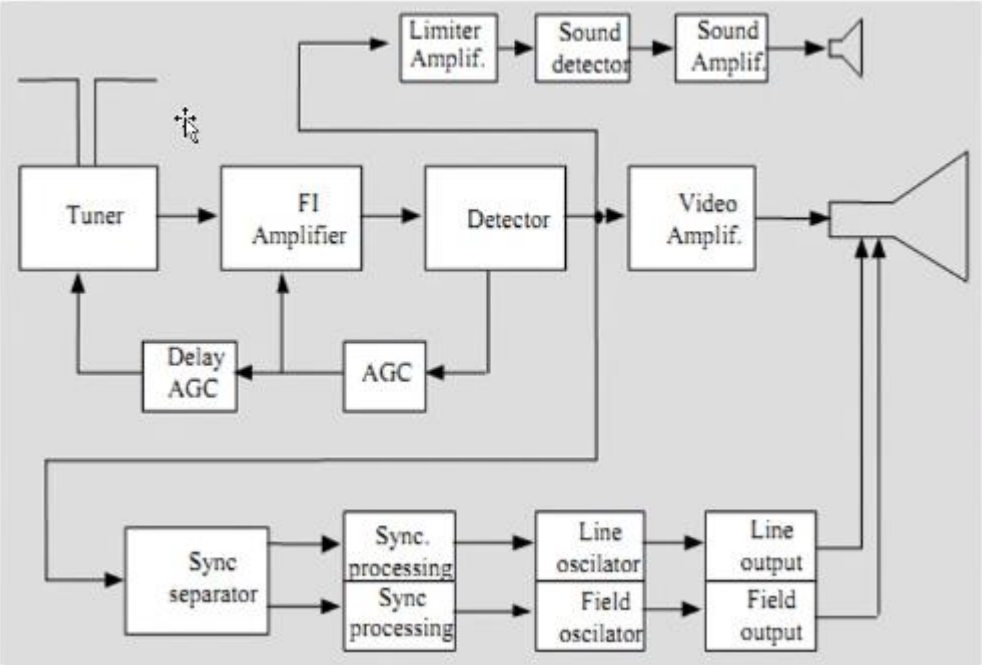
❖Basic Monochrome Television System



▪ TV transmitter - B/W



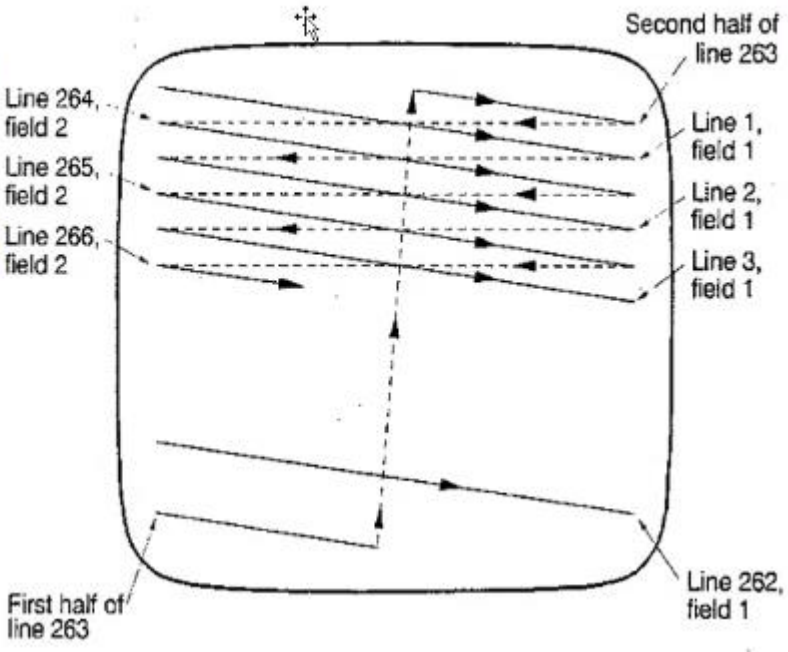
▪ TV receiver - B/W



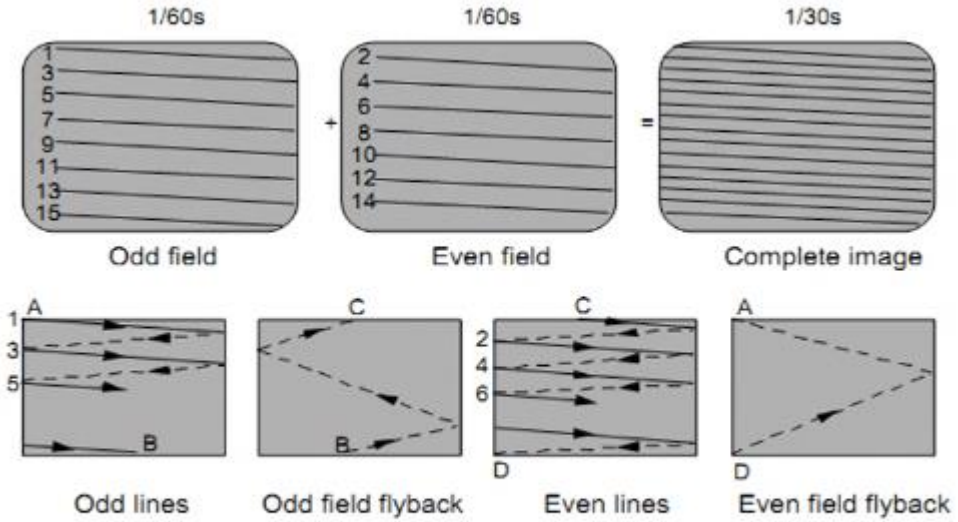
▪ TV scanning principle-Interlaced Scanning

A frame is divided into two fields to save bandwidth.

It takes advantages of the image retention of our eyes



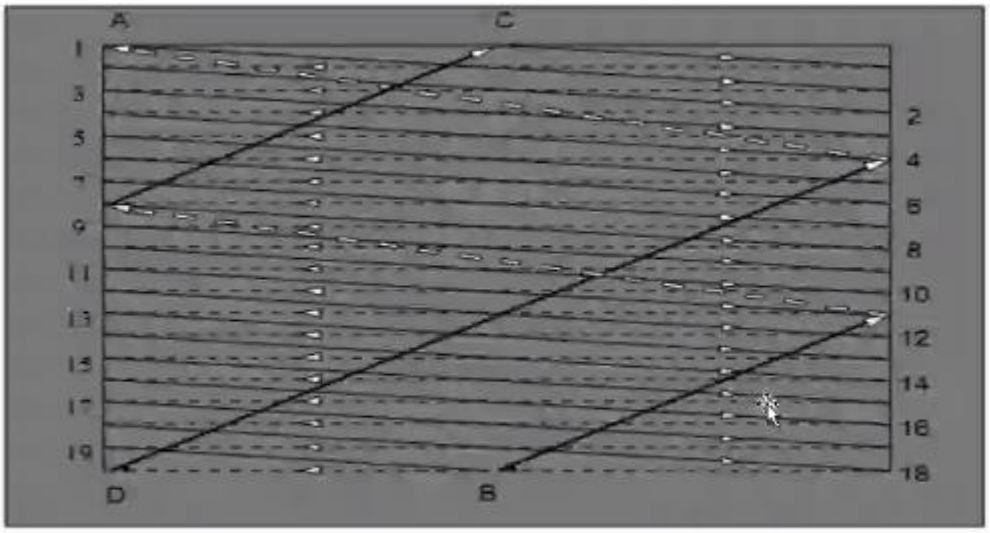
▪ Interlaced TV system



Number of lines: 525
Number of images: 30

Number of fields: 60
2 fields = 1 image

▪ Interlaced TV system



Horizontal scanning
Horizontal flyback
Vertical flyback

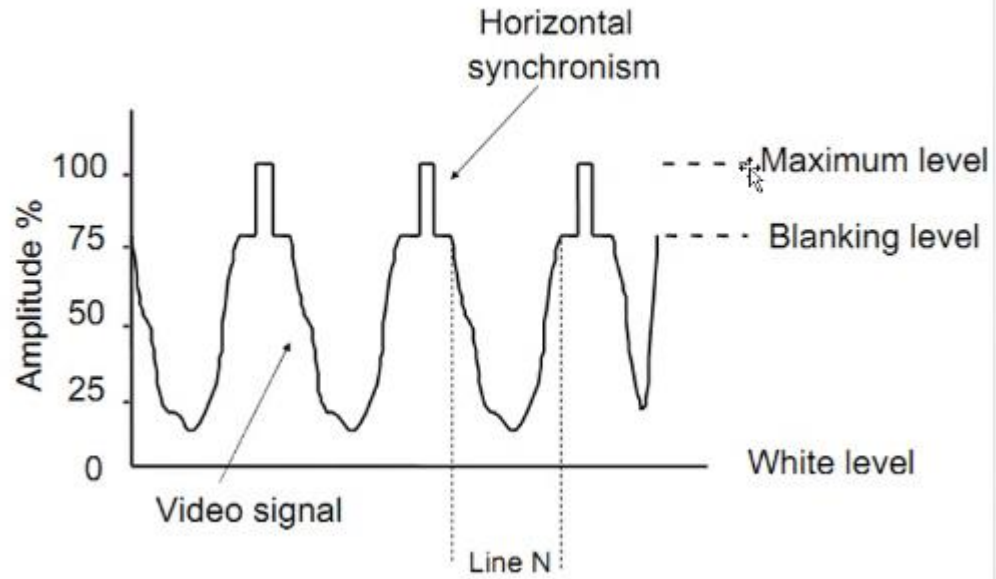
❖ Scanning Standards

	Color type	Fields per second	Frames per second	Lines per frame	Lines per second
United States					
Monochrome		60	30	525	15,750
Color	NTSC	59.94	29.97	525	15,734
England					
Monochrome		50	25	405	10,126
Color	PAL	50	25	625	15,625
Japan					
Color	NTSC	59.94	29.97	525	~~17,734~~ 15,734
France					
Monochrome		50	25	819	20,475
Color	SECAM	50	25	625	15,625
Germany					
Color	PAL	50	25	625	15,625
Former USSR					
Color	SECAM	50	25	625	15,625

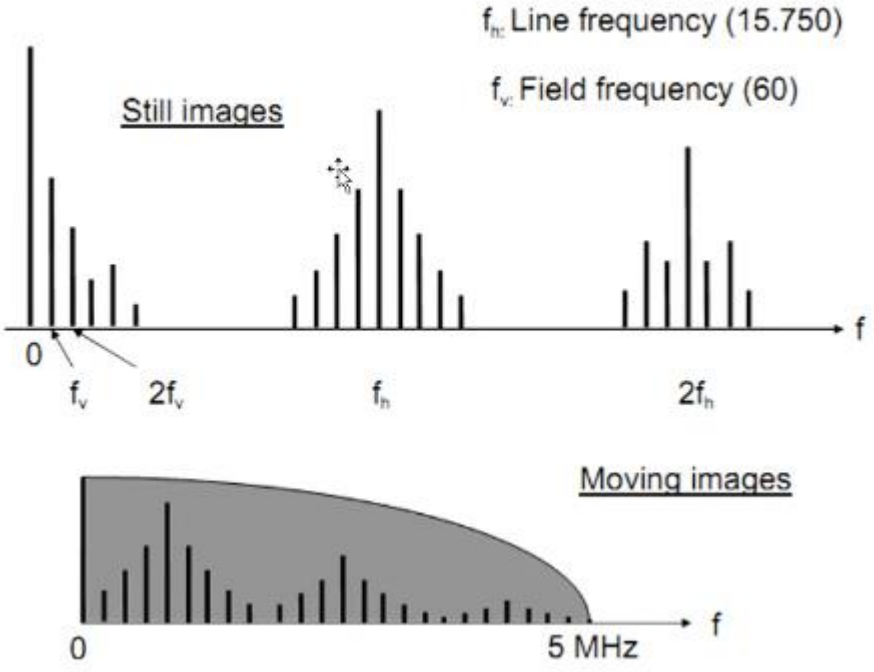
Color Television System Bandwidths

Country	System	Bandwidth, MHz
United States, Japan	NTSC	4.2
Canada, Mexico	NTSC	4.2
Great Britain	PAL	5.5
Germany, Austria, Italy	PAL	5.0
France	SECAM	6.0
Former USSR	SECAM	6.0

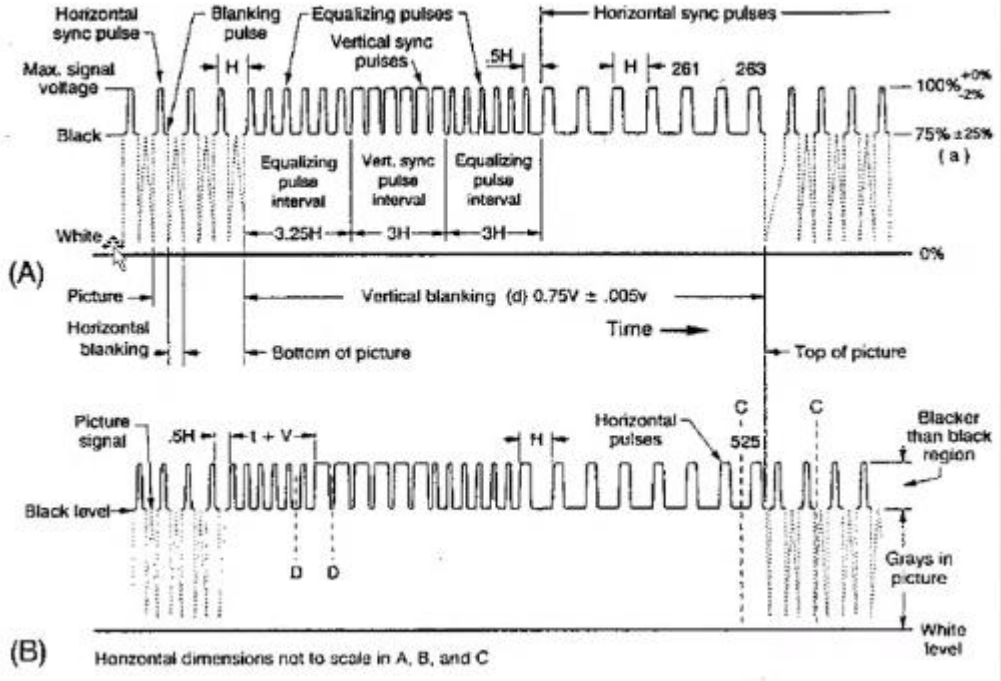
❖ TV signal - B/W



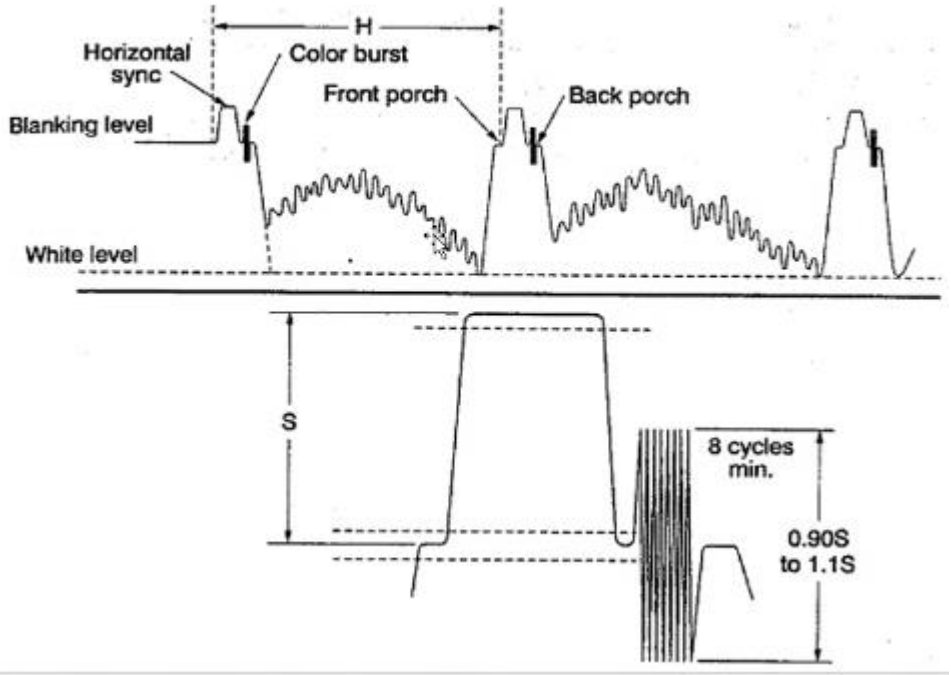
❖ TV signal spectrum - B/W



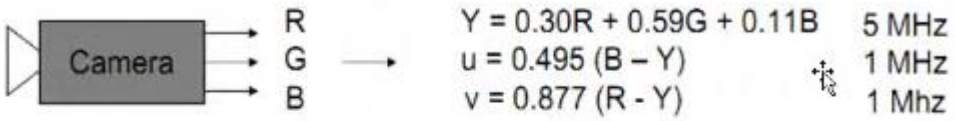
❖Luminance Video Signal Waveform



❖ Chrominance Signal Components



❖ Component TV color signals



Bandwidth is 7 Mhz



Y - Luminance u = 0.495 (B – Y) v = 0.877 (R - Y)

Paper Summary Format

- Include title of paper, along with publishing information (see example)
- Email summaries to your supervisor
- Include copy of paper in your email

Summary:

- A quick description of the paper and its results/experimental procedures
- Do not *discuss* the results, only summarize

Insight:

- Discuss the results/procedures and how they are exciting
- Answer the question: Why was this paper published?
- You can also include if you do not agree with any of the results, or methods

Application to Research:

- How do the results/procedures apply to your project
- What do you take away from the paper?
- How does it help you?